Data Preprocessing

Cleaning, integration, reduction, transformation and discretization

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Source: J. Han, M. Kamber & J. Pei, Data Mining: Concepts and Techniques, 3rd ed., Morgan Kaufmann, 2011 — Chapter 3

Roadmap

1. Why preprocess? Data quality
2. Data cleaning
3. Data integration
4. Data reduction
5. Data transformation and discretization
6. Summary and check-your-understanding

Learning outcomes

- Diagnose data-quality problems and choose a remedy
- Handle missing values and smooth noisy data
- Detect redundancy with χ^2, correlation and covariance
- Reduce data by dimensionality and numerosity reduction
- Normalize, discretize and build concept hierarchies

Why Preprocess? Data Quality

Section 1

Garbage in, garbage out

Real-world databases are incomplete, noisy and inconsistent — large, drawn from heterogeneous sources, and entered by people and devices that make mistakes.

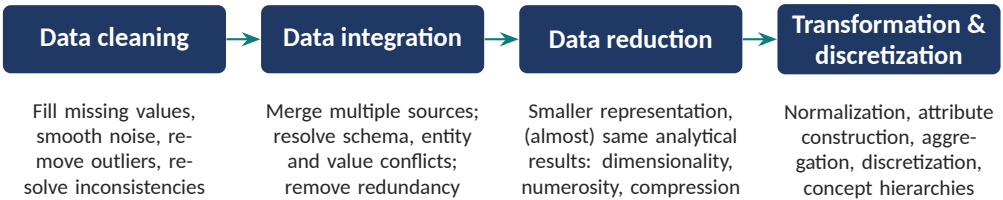
Six dimensions of data quality

- **Accuracy** — correct values
- **Completeness** — no missing values or attributes
- **Consistency** — no contradictions (codes, formats)
- **Timeliness** — up to date when needed
- **Believability** — trusted by users
- **Interpretability** — easily understood

Typical causes

- Faulty instruments, transmission errors
- Human or software entry errors (default values such as 1 January for birth date)
- Data not captured, or deleted as inconsistent
- Inconsistent naming and units across sources

Four major preprocessing tasks



Key idea The tasks are not strictly sequential: removing redundancy is also a form of reduction, and discretization also reduces data. In practice preprocessing is **iterative** and often consumes the majority of a data-mining project's effort.

Data Cleaning

Section 2

Handling missing values

1. **Ignore the tuple** — usual when the class label is missing; wasteful when many attributes are missing.
2. **Fill in manually** — accurate but infeasible for large data.
3. **Global constant** (e.g. “Unknown”, $-\infty$) — simple, but the miner may treat “Unknown” as a meaningful concept.
4. **Measure of central tendency** — mean for symmetric data, median for skewed data.
5. **Class-wise mean or median** — use only tuples of the same class (e.g. same credit-risk category).
6. **Most probable value** — predict with regression, Bayesian inference or a decision tree trained on the other attributes.

Key idea Methods 3–6 **bias** the data, as the filled value may be wrong. Method 6 is the most popular because it uses the most information. Note: a missing value is not always an error — “driver’s licence number” is legitimately blank for non-drivers.

Worked Example 1 — filling missing income

Customer	Risk class	Income (Tk '000)
C1	low	60
C2	low	70
C3	low	?
C4	high	25
C5	high	35
C6	high	?
C7	low	80

- **Global mean:** $(60 + 70 + 25 + 35 + 80)/5 = \mathbf{54}$ for both C3 and C6
- **Class-wise mean:** low $= (60 + 70 + 80)/3 = \mathbf{70}$ for C3; high $= (25 + 35)/2 = \mathbf{30}$ for C6

Key idea The global mean gives a high-risk customer an income nearly twice that of other high-risk customers; the class-wise mean respects the structure.

Noisy data: smoothing by binning

Noise is random error or variance in a measured variable. **Binning** smooths sorted values by consulting their neighbourhood:

- Partition the sorted data into bins (**equal-frequency** or **equal-width**)
- **Smoothing by bin means**: replace each value by its bin mean
- **Smoothing by bin medians**: replace by the bin median
- **Smoothing by bin boundaries**: replace by the closer of the bin's min and max

> **Key idea** Binning performs **local** smoothing. Larger bins give stronger smoothing. Binning is also a discretization technique (Section 5).

Worked Example 2 — binning

Sorted prices (Tk '00): 6, 8, 13, 18, 20, 25, 26, 30, 37 — equal-frequency bins of size 3

	Bin 1	Bin 2	Bin 3
Partition	6, 8, 13	18, 20, 25	26, 30, 37
By means	9, 9, 9	21, 21, 21	31, 31, 31
By medians	8, 8, 8	20, 20, 20	30, 30, 30
By boundaries	6, 6, 13	18, 18, 25	26, 26, 37

Boundaries: 8 is closer to 6 than to 13; 20 is closer to 18; 30 is closer to 26.

Equal-width alternative (width ≈ 10.3): [6, 16.3) $\rightarrow$ 6, 8, 13; [16.3, 26.7) $\rightarrow$ 18, 20, 25, 26; [26.7, 37] $\rightarrow$ 30, 37 — bins of unequal size.

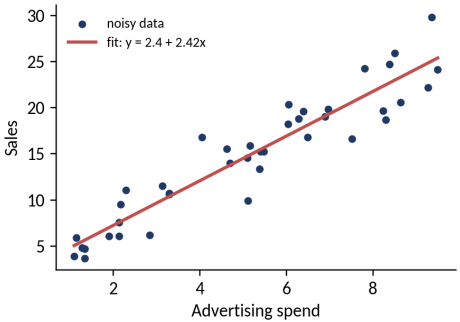
Other smoothing techniques

Regression

- Fit a function (linear or multiple regression) and replace values by fitted values
- One attribute is used to predict another

Outlier analysis

- Clustering: values falling outside clusters are candidate outliers
- Statistical tests and density methods (Chapter 12)



Data cleaning as a process

1. Discrepancy detection

- Use **metadata**: domain, type, acceptable range, dependencies
- Check basic statistics (Chapter 2): values $> 2\sigma$ from the mean, inconsistent codes, formats (“2026/10/05” vs “05-10-26”)
- **Unique rule**: each value distinct; **consecutive rule**: no gaps between min and max; **null rule**: how blanks and “?” are coded

2. Data transformation

- **Data scrubbing** tools: domain knowledge (postal addresses, spell checking)
- **Data auditing** tools: rule and relationship discovery
- **ETL** tools and **migration** tools

Key idea The two steps iterate: fixing one discrepancy can reveal others. Interactive tools such as *Potter’s Wheel* integrate detection and transformation with immediate feedback.

Data Integration

Section 3

Merging data from multiple sources

- **Entity identification problem**: is `customer_id` in one database the same as `cust_number` in another? Use metadata — name, meaning, type, range, null rules — and watch for functional dependencies and referential constraints.
- **Redundancy**: an attribute may be derivable from others (annual revenue from monthly revenue), or names may be inconsistent. Detect with **correlation analysis**.
- **Tuple duplication**: denormalized tables and data-entry errors produce duplicate or inconsistent tuples (same customer, two addresses).
- **Data value conflicts**: the same entity with different values — units (kg vs lb), currency, scale (hotel prices with or without tax), abstraction level (branch vs region sales).

Correlation analysis for nominal data: the χ^2 test

For attributes A (c values) and B (r values) in a contingency table:

$$\chi^2 = \sum_{i=1}^{c} \sum_{j=1}^{r} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}, \qquad e_{ij} = \frac{\text{count}(A = a_i) \times \text{count}(B = b_j)}{n}$$

- Tests the hypothesis that A and B are **independent**, with $(r-1)(c-1)$ degrees of freedom
- Cells contributing most are those whose observed count is far from the expected count
- Reject independence if χ^2 exceeds the critical value

Key idea Correlation does not imply causality.

Worked Example 3 — χ^2 test

200 students: programme vs preferred exam mode

	Online	Offline	Total
CSE	60 (45)	40 (55)	100
EEE	30 (45)	70 (55)	100
Total	90	110	200

Expected counts in parentheses, e.g.

$e_{11} = 100 \times 90/200 = 45$

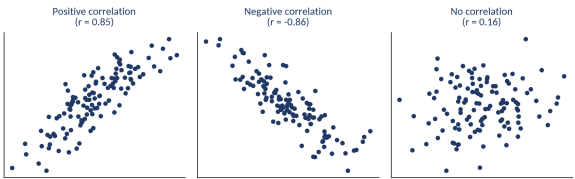
$$\chi^2 = \frac{15^2}{45} + \frac{15^2}{55} + \frac{15^2}{45} + \frac{15^2}{55} = \mathbf{18.18}$$

- df $= (2-1)(2-1) = 1$
- $\chi^2_1(0.001) = 10.83$
- $18.18 > 10.83 \Rightarrow$ reject independence: programme and exam preference are **correlated**

Correlation for numeric data: Pearson's coefficient

$$r_{A,B} = \frac{\sum_{i=1}^{n}(a_i - \bar{A})(b_i - \bar{B})}{n\,\sigma_A\,\sigma_B} = \frac{\sum_{i=1}^{n} a_i b_i - n\bar{A}\bar{B}}{n\,\sigma_A\,\sigma_B}, \qquad -1 \leq r_{A,B} \leq +1$$

- $r > 0$: positively correlated; the higher the value, the stronger
- $r = 0$: no *linear* relationship
- $r < 0$: negatively correlated
- A high $|r|$ suggests one attribute may be **redundant**



Covariance and Worked Example 4

$$\mathrm{Cov}(A,B) \;=\; \frac{1}{n}\sum_{i=1}^{n}(a_i-\bar{A})(b_i-\bar{B}) \;=\; E(A\cdot B)-\bar{A}\bar{B} \qquad r_{A,B} \;=\; \frac{\mathrm{Cov}(A,B)}{\sigma_A\sigma_B}$$

Weekly closing prices of two stocks over five weeks

$A=(12,15,11,18,14)$, $\bar{A}=14$; $\quad B=(30,36,28,40,31)$, $\bar{B}=33$

Deviation products: $6+3+15+28+0=52 \Rightarrow \mathrm{Cov}(A,B)=52/5=\mathbf{10.4}>0$: the stocks tend to rise together ($r=0.97$).

Hours studied $(2,4,5,7,8)$ vs marks $(50,60,65,80,85)$: $r=137/137.56=\mathbf{0.996}$.

Independence $\Rightarrow \mathrm{Cov}=0$, but $\mathrm{Cov}=0$ does *not* imply independence (unless the data are multivariate normal).

Data Reduction

Section 4

Data reduction strategies

Obtain a reduced representation that is much smaller yet yields (almost) the same analytical results. Time spent reducing must not outweigh time saved in mining.

Dimensionality reduction

Fewer attributes: wavelet transforms, PCA, attribute subset selection

Numerosity reduction

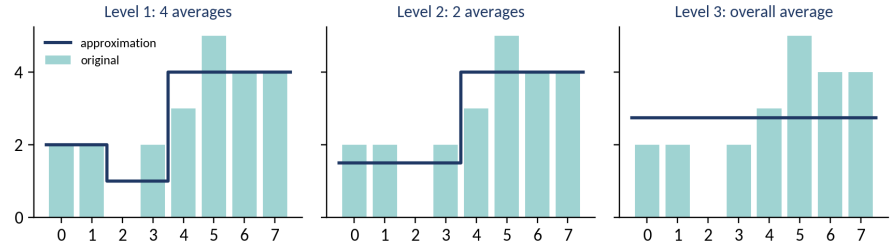
Smaller representations of the data: parametric (regression, log-linear) and non-parametric (histograms, clustering, sampling, data cubes)

Data compression

Lossless: original reconstructable
Lossy: approximation only (wavelets, PCA)

Wavelet transforms

- The **discrete wavelet transform** (DWT) turns a data vector X into an equal-length vector X' of wavelet coefficients
- Keep only the strongest coefficients (set the rest to 0) $\Rightarrow$ sparse, lossy compression; inverse DWT reconstructs an approximation
- Removes noise without smoothing away main features; works on sparse, skewed and ordered data; complexity $O(n)$
- Pyramid algorithm: pairwise **smoothing** (averages) and **differencing** (details), halving length each level; data length padded to a power of 2



Worked Example 5 — Haar wavelet transform

$X = (2, 2, 0, 2, 3, 5, 4, 4)$ **averages** $(a+b)/2$**, details** $(a-b)/2$

Level 1	averages $(2,\ 1,\ 4,\ 4)$	details $(0,\ -1,\ -1,\ 0)$
Level 2	averages $(1.5,\ 4)$	details $(0.5,\ 0)$
Level 3	average (2.75)	detail (-1.25)

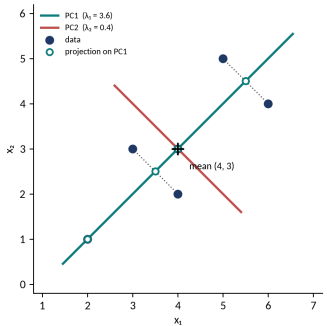
Coefficient vector: $X' = (2.75,\ -1.25,\ 0.5,\ 0,\ 0,\ -1,\ -1,\ 0)$

Key idea Three of the eight coefficients are already zero. Zeroing $|c| \leq 0.5$ keeps only 4 values; reconstruction is approximate but preserves the overall shape. (Normalized Haar uses factors of $1/\sqrt{2}$; the averaging form is easier by hand.)

Principal component analysis (recap)

1. Normalize the data (each attribute in the same range)
2. Compute *k* orthonormal vectors — the **principal components** — the eigenvectors of the covariance matrix
3. Sort by decreasing variance (eigenvalue)
4. Drop weak components; project the data onto the strongest

Handles sparse and skewed data; often reveals hidden relationships. Numeric attributes only. See the PCA lecture for the full derivation.



Attribute subset selection

Find a minimal attribute set whose class distribution is as close as possible to that of all attributes. d attributes $\Rightarrow 2^d$ subsets, so use greedy heuristics with tests of significance or information gain.

Method	Illustration with $\{A_1, \ldots, A_6\}$
Stepwise forward selection	$\{\} \to \{A_1\} \to \{A_1, A_4\} \to \{A_1, A_4, A_6\}$
Stepwise backward elimination	$\{A_1, \ldots, A_6\} \to \{A_1, A_3, A_4, A_5, A_6\} \to \{A_1, A_4, A_5, A_6\} \to \{A_1, A_4, A_6\}$
Forward + backward combined	add the best and remove the worst at each step
Decision tree induction	attributes not appearing in the tree are irrelevant

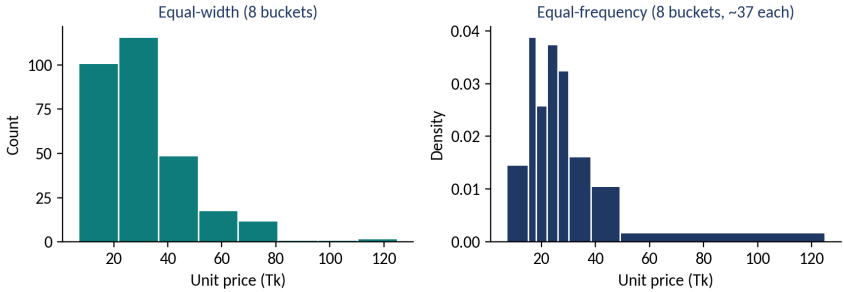
Key idea Attribute **construction** can also help: e.g. *area* from *height* $\times$ *width*.

Parametric numerosity reduction

- **Linear regression**: $y = wx + b$; store only w and b (least squares). **Multiple regression** extends to several predictors.
- **Log-linear models**: approximate discrete multidimensional probability distributions from lower-dimensional combinations — estimate each cell of a base cuboid from its lower-order cuboids.
- Both handle skewed data; regression copes well with sparse data; log-linear models scale to about 10 dimensions.

> **Key idea** Store the *model parameters* instead of the data, plus possibly outliers that the model fits poorly.

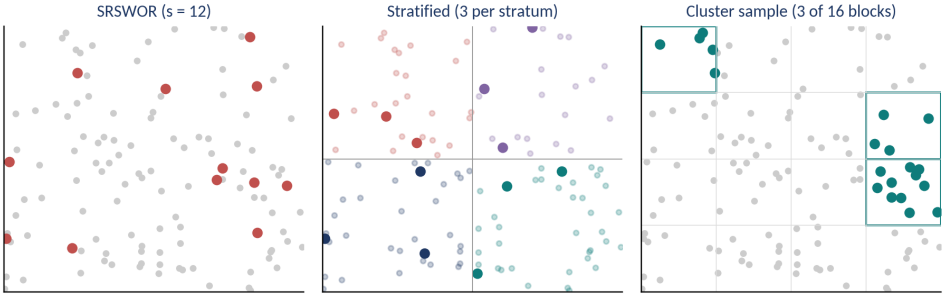
Histograms and clustering



- **Equal-width**: buckets of uniform range; **equal-frequency**: about the same count per bucket
- Histograms are effective for sparse, dense, highly skewed and uniform data; multidimensional histograms capture dependencies
- **Clustering**: store cluster representations (centroid, diameter) in place of the data — effective when data are naturally clustered, poor for smeared data

Sampling

A large data set D of N tuples is represented by a random sample of size $s \ll N$; cost is proportional to s, not N.



- **SRSWOR** / **SRSWR**: simple random sampling without / with replacement (each tuple equally likely)
- **Cluster sample**: tuples grouped into M disjoint clusters (e.g. disk pages); draw an SRS of clusters
- **Stratified sample**: SRS within each stratum — guarantees representation of rare groups (skewed data)

Data cube aggregation

Quarter (2025)	Sales (Tk lakh)
Q1	224
Q2	408
Q3	350
Q4	586
Annual	**1,568**

- Aggregate quarterly to annual if the analysis concerns annual sales — smaller, without loss for that task
- A **data cube** stores multidimensional aggregates; the **base cuboid** is the lowest level, the **apex cuboid** the highest
- Use the smallest cuboid relevant to the query

Transformation and Discretization

Section 5

Transformation strategies

1. **Smoothing** — remove noise (binning, regression, clustering)
2. **Attribute construction** — add new attributes from existing ones
3. **Aggregation** — summarize (daily $\rightarrow$ monthly sales); builds data cubes
4. **Normalization** — scale into a small range such as $[0, 1]$ or $[-1, 1]$
5. **Discretization** — replace raw numeric values by interval labels (0–10, 11–20) or concept labels (youth, adult, senior)
6. **Concept hierarchy generation for nominal data** — *street* generalized to *city* or *country*

Normalization

Gives every attribute equal weight — essential for distance-based methods (*k*-NN, clustering) and helps neural-network training converge.

Min-max: $v'_i = \dfrac{v_i - \min_A}{\max_A - \min_A}(\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$

z-score: $v'_i = \dfrac{v_i - \bar{A}}{\sigma_A}$ (or use the mean absolute deviation s_A for robustness)

Decimal scaling: $v'_i = \dfrac{v_i}{10^j}$, j = smallest integer with $\max|v'_i| < 1$

Min-max preserves relationships but raises an "out-of-bounds" error if future data fall outside $[\min_A, \max_A]$. Store the parameters to normalize future data consistently.

Worked Example 6 — normalization

Monthly income: min Tk 15,000, max Tk 1,20,000, mean Tk 45,000, s.d. Tk 15,000

Min-max to $[0, 1]$ for Tk 50,000:
$(50{,}000 - 15{,}000)/(1{,}20{,}000 - 15{,}000) = 35{,}000/1{,}05{,}000 = \mathbf{0.333}$

z-score for Tk 50,000: $(50{,}000 - 45{,}000)/15{,}000 = \mathbf{0.333}$

Decimal scaling: values range from -850 to 920; max $|v| = 920 \Rightarrow j = 3$, so $920 \to \mathbf{0.920}$, $-850 \to \mathbf{-0.850}$

Key idea The two 0.333s coincide only by chance. A z-score is unbounded and measures distance from the mean in standard deviations; min-max maps the observed range exactly onto $[0, 1]$.

Discretization

Classified by

- **Supervised** (uses class labels) vs **unsupervised**
- **Top-down** (splitting) vs **bottom-up** (merging)
- Applied recursively, discretization builds a **concept hierarchy** for a numeric attribute

Methods

- **Binning**: equal-width or equal-frequency; unsupervised, top-down
- **Histogram analysis**: unsupervised; recursive partitioning with a minimum interval size
- **Cluster analysis**: top-down or bottom-up
- **Decision tree** (entropy): supervised, top-down — choose the split point minimizing class entropy
- **Correlation** (ChiMerge): supervised, bottom-up — merge adjacent intervals with the lowest χ^2

Concept hierarchies for nominal data

1. **Partial ordering specified by experts** at schema level: *street* $<$ *city* $<$ *district* $<$ *division* $<$ *country*
2. **Explicit data grouping** for a portion of the hierarchy: {Savar, Dhamrai} $\subset$ Dhaka district
3. **Set of attributes only**: order automatically by the number of distinct values — fewest at the top
4. **Partial set**: embed semantics so that specifying one attribute (e.g. *city*) drags in the rest



Distinct-value heuristic (Bangladesh)

The heuristic can fail: *year* (20 values) $<$ *month* (12) $<$ *day of week* (7) is wrong.

Summary

Section 6

Take-home messages

- Data quality has six facets: accuracy, completeness, consistency, timeliness, believability, interpretability.
- **Cleaning**: fill missing values (prefer model-based or class-wise estimates), smooth noise, treat cleaning as an iterative detect-and-transform process.
- **Integration**: resolve entities and value conflicts; detect redundancy with χ^2 (nominal) and correlation or covariance (numeric).
- **Reduction**: wavelets, PCA and subset selection reduce dimensions; regression, histograms, clustering, sampling and cubes reduce volume.
- **Transformation**: normalize (min-max, z-score, decimal scaling), discretize, and generate concept hierarchies.

Check your understanding

1. Smooth 4, 10, 11, 15, 21, 24, 25, 31, 36 by bin means and by bin boundaries (equal-frequency bins of 3).
2. For 100 customers, 30 of 40 smokers and 20 of 60 non-smokers bought insurance. Compute χ^2; are the attributes independent at $\alpha = 0.001$?
3. Normalize 600 in a range 200–1000 to $[0, 1]$; give its z-score if mean $= 500$, s.d. $= 200$.
4. Give the Haar coefficients of $(4, 6, 10, 12)$.
5. Why is stratified sampling preferred to SRS for highly skewed class distributions?

Answers: (1) means 8.33, 20, 30.67; boundaries 4, 11, 11 | 15, 24, 24 | 25, 36, 36. (2) $\chi^2 = 16.67 > 10.83$: dependent. (3) 0.5; 0.5. (4) $(8, -3, -1, -1)$.

Further reading

- J. Han, M. Kamber and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed., Morgan Kaufmann, 2011 — Chapter 3.
- D. Pyle, *Data Preparation for Data Mining*, Morgan Kaufmann, 1999.
- V. Raman and J. M. Hellerstein, “Potter’s Wheel: an interactive data cleaning system,” *Proc. VLDB*, 2001.
- R. Kerber, “ChiMerge: discretization of numeric attributes,” *Proc. AAAI*, 1992.

Thank you

Questions and discussion

Asif Zaman · Founder, Corpusian